Sadguru Balumama Shikshan Prasarak Mandal's

K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

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Internships Daily Diary(2025-26)

Department Of Artificial Intelligence & Machine Learning Engineering

Program Code :AI5K

Name of the Student : Sanika Dhanaji Chougale

Name of the Mentor (Faculty) : Mrs. P. M. Kumbhar

Name of the Mentor (Industry): Mr. Abhijit Gatade

Enrollment Number : 23213500044

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week 6	Mon, Date 07/07/25	♦ Introduction Machine Learning Concept	Read about ML LifeCycle	
	Tue, Date 08/07/25	♦ Types of ML	Understand different and write short notes	
	Wed, Date 09/07/25	♦ Overview of ML workflow	Study ML pipeline steps	
	Thu, Date 10/07/25	♦ Introduction to datasets & features Types	Learning about dataset formats & data importance	
	Fri, Date 11/07/25	♦ Common ML algorithms & use Cases	Make a chart of algorithms and suitable problem types.	
	Sat, Date 12/07/25	♦ Tools & Libraries used in ML	Install tools, explore sample datasets & basic ML Scripts	

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Internships Daily Diary (2025-26)

Week No: 06 **Day:** Monday **Date:** 07/07/25

• Introduction to Machine Learning Concepts

Today, I was introduced to the basic Concept of Machine Learning the beginning of our Machine Learning journey.

Where I learned that ML enables Systems to learn from data and make predictions or decisions without being explicity programmed.

The session focused on the ML Lifecycle, Which includes stage like data Collection, preprocessing, model training, evalution, and deployment. we discussed how ML is applied in various fields such as healthcare, finance, e-commerce and robotics.

This introductory session helped establish a strong conceptual foundation to build upon in the coming days.

Internships Daily Diary (2025-26)

Week No: 06 **Day:** Tuesday **Date:** 08/07/25

⊙ Types of Machine Learning

The focus of today's session was to understand the different types of Machine Learning, are follow:

- Supervised Learning – Where models are trained on labeled datasets
  e.g. (spam detection , Salary prediction).

- Unsupervised Learning – Where models analyze and cluster data without labeled output. (e.g. Customer Menagement)

- Reinforcement Learning – Where an agent learns to make decisions by interacting with an environment and receiving feedback.
  eg. Self – driving cars, game AI.

Internships Daily Diary (2025-26)

Week No: 06 **Day:** Wednesday **Date:** 09/07/25

Overview Of Machine Learning Workflow

Today's session involved a step-by-step overview of the Machine learning workflow, which describe how ML projects progress from raw data to deployment. The major stages include:

1] Defining the problem statement

2] Collecting and exploring the dataset

3] Data preprocessing and cleaning.

4] Feature engineering.

5] Model Selection and training

6] Model evalution

7] Hyperparameter tuning

8] model deployment and monitoring.

Internships Daily Diary (2025-26)

Week No: 06 **Day:** Thursday **Date:** 10/07/25

- **Introduction to Datasets and feature Types**

Todays, I learned about datasets, Which are at the core of any ML model. The session emphasized the formats Commonly used (CSV, JSON, Excel) and discussed the importance of well-structured data.

We also discussed different features types including:

- Numerical (e.g. age, salary)
- Categorical (e.g. gender, department)
- Boolean (true/false values)
- Textual (Comments, reviews)

Understanding how to handle each feature type is essential for Successful model training.

Internships Daily Diary (2025-26)

Week No: 06 **Day:** friday **Date:** 11/07/25

- Common Machine Learning Algorithms & Uses Cases:

Today, I was introduced to popular Machine Learning alggorithms and their respective use cases. The Session Convered:

- Linear Regression for numerical predictions
- Logistic Regression for binary classification
- Decision Trees and Rodom forest for both classification and regression.
- Support Vector Machines (SVM) for high-dimensional classification problems.
- K- Nearest Neighbor (KNN) for simple classification and regression.

Internships Daily Diary (2025-26)

Week No: 06 **Day:** Saturday **Date:** 12/07/25

- Tools and Libraries Used in ML

In the final session of the week, I explored the key tools and python libraries used in ML development These included:

- Numpy - for numrical Computations.
- Pandas - for data manipulation and analysis.
- Matplotlib & Seaborn - for data Visualization
- Scikit-learn - for implementing ML algorithms.
- Jupyter Notebook - as an interactive development environment

Commbnly used in real-world ML projects.